# GlucoTwin — system architecture

A patient-specific type 2 diabetes digital twin. Four input streams are harmonised onto one causal timeline, fused by the model layer, calibrated into prediction intervals and adverse-event probabilities, and served to two clinician interfaces. Raw patient records never leave the clinic.

## 1 · SOURCES

Inside the clinic — raw records never leave

STATIC EHR

### Health record

Age, sex, BMI, HbA1c, fasting glucose, eGFR, diagnoses, medications, genetic markers. Synthea-ready.

REAL · OPEN

### CGM · 145 patients

CGMacros (45 adults, PhysioNet) and Shanghai T2DM (100 patients, figshare). Glucose every 5 minutes over 10–14 day wears.

REAL + SIMULATED

### Wearable

Fitbit heart rate and activity (METs) from CGMacros. Sleep stages, HRV and step count from the simulator, labelled as such.

REAL + PRESETS

### Meals

Logged carbohydrate, protein, fat and fibre, plus 16 Indian meal presets with composition from IFCT-2017.

## 2 · HARMONISE & FUSE

### One causal timeline

**Common grid.** Every cohort resampled to a single 5-minute schema with a shared column set; loader checksums verified.

**Causal features only.** No feature may use information from after the prediction instant — enforced in code and covered by tests.

**Patient-grouped splits.** Training, inner-validation, calibration and holdout patients never overlap, so no patient appears on both sides of an evaluation.

**Gap and quality handling.** Sensor gaps flagged rather than silently interpolated; a per-window quality score travels with the forecast.

**Fused feature set.** Recent CGM history and its slope · time of day · meal macronutrients · heart rate and activity · the static profile.

`src/data · src/features`

## 3 · MODEL LAYER

### Fusion forecaster

Tuned LightGBM predicting the *change* from the current value at +30, +60 and +120 min. The served model, selected on inner-validation patients only.

### Mechanistic twin

A minimal glucose–insulin ODE fitted per patient. RK4 integration, extended Kalman filter for the state. Used for what-ifs and as a feature source.

### Personalisation

The population model plus fine-tuning, or a recursive-least-squares correction, on the patient's own first days. Cold-start curve measured per horizon.

### Federated training

FedAvg and FedProx across simulated clinics, with a differential-privacy noise sweep. Only model updates cross the clinic boundary.

## 4 · CALIBRATE

### 80% prediction intervals

Split-conformal and CV+. Measured coverage 80.8% on CGMacros and 79.5% on Shanghai at +60 min, against an 80% target. Coverage is also reported per subgroup, not only pooled.

### Event probabilities

Platt and isotonic calibration for crossing 180 mg/dL, 140 mg/dL and falling below 70 mg/dL. Reported as calibrated probabilities, never as bare alarm labels.

### Alert policy & explainability

Alert-burden curves: episodes caught per false alarm per day, so a clinic sets its own threshold. SHAP attribution over the fused feature set explains each forecast.

## 5 · CLINICAL OUTPUTS

### Forecast + band

Trajectory to +120 min with its 80% interval and one plain-language sentence a clinician can read instead of the chart.

### Adverse-event risk

Calibrated chance of hyperglycaemia or hypoglycaemia inside the window, with the alert threshold the clinic chose.

### What-if answers

Counterfactual meal size, carbohydrate, timing, an Indian preset swap, and a post-meal walk — through the monotone model and the twin.

### Trust surface

Sensor data quality, personalisation depth, input tier available, feature attribution, and the limits printed beside each answer.

## 6 · INTERFACES

### Web console `web/`

A single-page clinician console. Patient list with triage ordering, then per patient: forecast, what-if lab, glucose report (AGP, time in range, GMI per the 2019 consensus), health record, sleep and activity context, visit summary, and an evidence section carrying every cohort-level result.

Role-gated access — clinician and care coordinator see different fields — with a session access log. Dark and light themes, mg/dL ⇄ mmol/L, readable at tablet width.

### Streamlit application `make app`

Ten pages: Overview, Live twin, Glucose report, What-if lab, Twin state, Personalization, Input tiers, Data quality, Evidence, and a PDF export for the patient file.

Both interfaces read the same bundle, so the two never disagree on a number. Every page carries the decision-support disclaimer: the clinician remains the decision maker and the system never recommends a dose or a drug.

## METHOD — HOW EVERY NUMBER WAS OBTAINED

- **Pre-registered.** Each protocol, candidate set and decision rule written to `docs/DECISIONS.md` before the run that tested it. Over 70 numbered entries.
- **Paired patient bootstrap.** Confidence intervals computed over patients, never over time windows — readings from one patient are not independent.
- **Untouched holdout.** Final claims confirmed on patients never used for any choice anywhere in the project; the runner refuses a second run.
- **Null results published.** Where a signal does not help, the number is reported at full size rather than dropped.

## EVALUATION PROTOCOL

- **Patient-grouped cross-validation** with repeats, an inner validation split for every choice, and a reserved holdout cohort.
- **Baselines it must beat:** persistence (assume glucose stays where it is), ridge regression, and a GRU sequence model.
- **Horizons:** +30, +60 and +120 minutes, scored on mean absolute error.
- **Beyond accuracy:** interval coverage, event AUROC, alert burden, what-if plausibility checks, and 87 matched real breakfasts.
- **Stress tests:** cross-cohort transfer and a synthetic Indian cohort at four insulin-resistance settings.

## MEASURED RESULTS

| METRIC | VALUE |
|---|---|
| MAE at +30 min | 11.6 |
| MAE at +60 min | 16.8 |
| MAE at +120 min | 21.2 |
| AUROC, above 180 | 0.82 |
| AUROC, below 70 | 0.81 |
| Interval coverage | 80.8% |
| Transfer gap | 3.4 |

## PRIVACY BY DESIGN

- Anonymised open or synthetic data only. No real patient record enters the repository, and none ever did.
- **Federated:** FedAvg matches pooled central training (difference 0.09 to 0.12 mg/dL, intervals include zero) and beats a single small clinic alone by about 2 mg/dL.
- Data minimisation is the design: records stay at the clinic, only model updates are exchanged. No formal differential-privacy guarantee is claimed at this clinic count.
- Role-gated access with a session log; handling mapped to India's **DPDP Act** in `docs/limitations.md`.

## ENGINEERING & SCOPE

- Python 3.11 · LightGBM · PyTorch · SHAP · Streamlit. Runs on a laptop CPU; no GPU required.
- One `make` target per stage; every figure and table regenerates from source. CI runs ruff, black, pytest and a smoke run on each push. Docker image, datasheet and model card included.
- One condition: **type 2 diabetes**. Decision support only — not a medical device.
- India, sleep and federated results are **simulations**: they test design, not clinical performance. Not prospectively validated.



Code MIT. Datasets keep their own licences. Simulated streams are labelled SIMULATED throughout.